

Ex.No. 14: 5G Throughput vs SNR with Modulation Labels

Objective 1 Matlab Code and Output:

```
clc;

clear;

close all;

% SNR values (dB)

SNR = 0:5:30;

% Maximum Throughput (Mbps)

Max_TP = 100;

% Throughput calculation

Throughput = Max_TP*(1-exp(-SNR/10));

disp('SNR(dB) Throughput(Mbps)')

disp([SNR' Throughput'])

figure

% ----- Graph 1 -----

subplot(2,1,1)

plot(SNR,Throughput,'-o','LineWidth',2)

title('5G Throughput vs SNR')

xlabel('Signal-to-Noise Ratio (dB)')

ylabel('Throughput (Mbps)')

grid on

% ----- Graph 2 -----

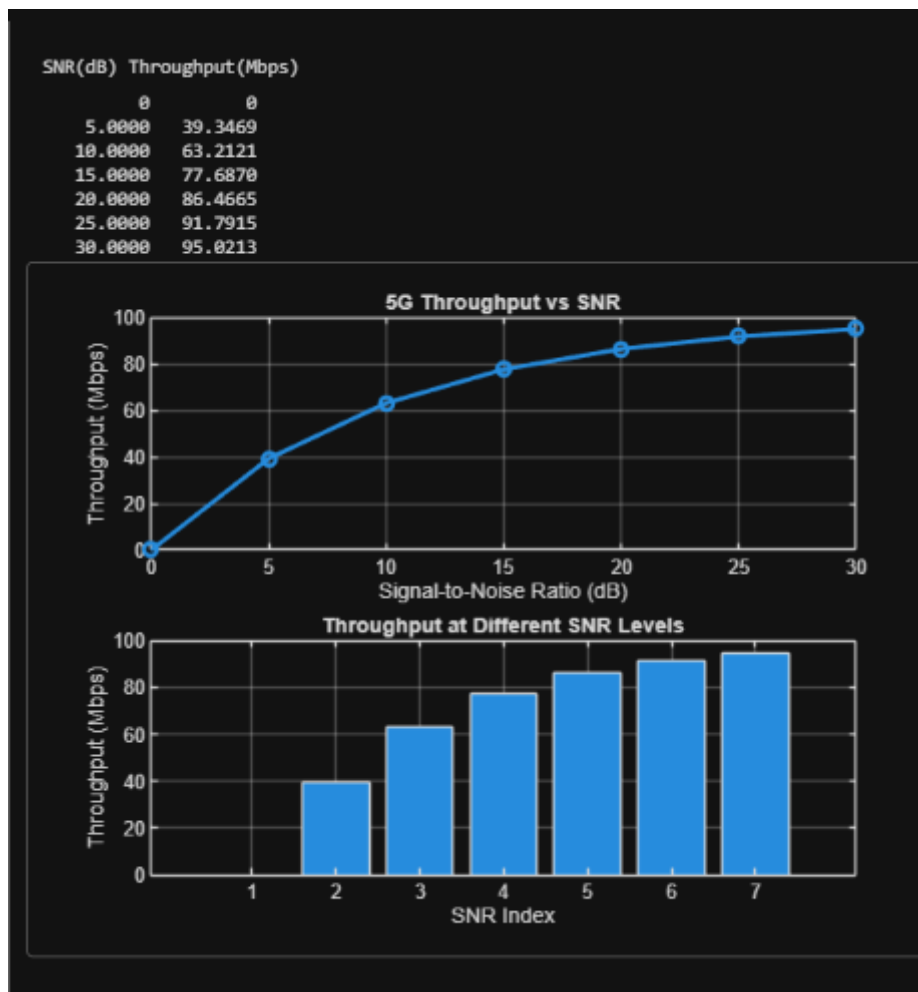
subplot(2,1,2)

bar(Throughput)

title('Throughput at Different SNR Levels')

xlabel('SNR Index')

ylabel('Throughput (Mbps)');grid on
```



Objective 2 Matlab Code and Output:

```

clc;

clear;

close all;

% SNR values (dB)

SNR = 0:5:30;

% Simulated Bit Error Rate (BER)

BER = [0.30 0.18 0.10 0.05 0.02 0.008 0.001];

% Reliability (%)

Reliability = (1-BER)*100;

disp('SNR(dB) BER Reliability(%)')

disp([SNR' BER' Reliability'])

figure

```

```
% ----- Graph 1 -----
```

```
subplot(2,1,1)
```

```
semilogy(SNR,BER,'-o','LineWidth',2)
```

```
title('SNR vs Bit Error Rate in 5G')
```

```
xlabel('Signal-to-Noise Ratio (dB)')
```

```
ylabel('Bit Error Rate (BER)')
```

```
grid on
```

```
% ----- Graph 2 -----
```

```
subplot(2,1,2)
```

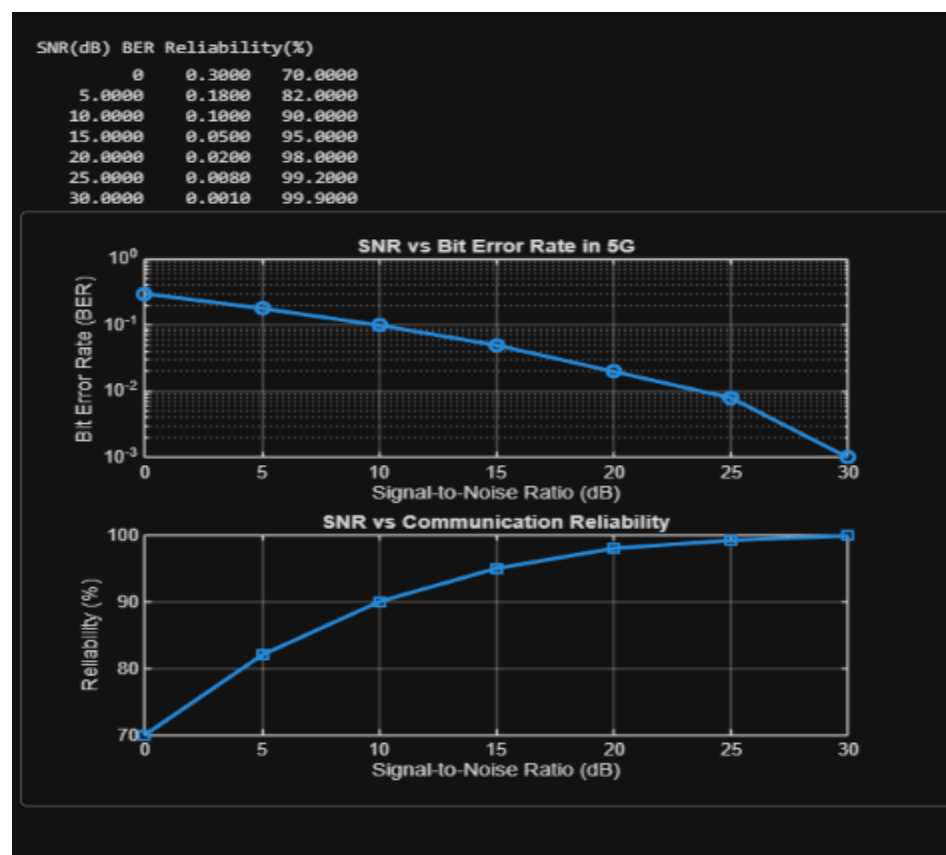
```
plot(SNR,Reliability,'-s','LineWidth',2)
```

```
title('SNR vs Communication Reliability')
```

```
xlabel('Signal-to-Noise Ratio (dB)')
```

```
ylabel('Reliability (%)')
```

```
grid on
```



Objective 3 Matlab Code and Output:

```
clc;

clear;

close all;

% Modulation schemes
Modulation = {'QPSK','16-QAM','64-QAM','256-QAM'};

% Required SNR (dB)
SNR = [5 10 18 25];

% Spectral Efficiency (bits/s/Hz)
Efficiency = [2 4 6 8];

disp('Required SNR (dB) and Spectral Efficiency')
disp(table(Modulation,'SNR',Efficiency', ...
'VariableNames',{'Modulation','SNR_dB','Efficiency'}))

figure

% ----- Graph 1 -----
subplot(2,1,2)

plot(SNR,Efficiency,'-o','LineWidth',2)

title('SNR vs Spectral Efficiency')

xlabel('Signal-to-Noise Ratio (dB)')

ylabel('Spectral Efficiency (bits/s/Hz)')

grid on

% ----- Graph 2 -----
subplot(2,1,1)

plot(1:4,SNR,'-s','LineWidth',2)

xticks(1:4)

xticklabels(Modulation)

title('Required SNR for Different Modulation Schemes')

xlabel('Modulation Scheme')

ylabel('Required SNR (dB)');grid on
```

Required SNR (dB) and Spectral Efficiency

Modulation	SNR_dB	Efficiency
{ 'QPSK' }	5	2
{ '16-QAM' }	10	4
{ '64-QAM' }	18	6
{ '256-QAM' }	25	8

